Week 7

Spring 2027 · Section 029 · MW 9:00–10:20 AM · RRC Room 8210

Course Level Objectives:

- Explain the role of probability in statistics. (CLO 4)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Classify a numerical summary as a parameter or a statistic. (CLO 5) **2.** Use the correct notation for parameters (μ, σ, p, ρ) and statistics ($\bar{x}$, s, $\hat{p}$, r). (CLO 5) **3.** Describe how a sampling distribution is built from repeated samples and what each dot represents. (CLO 5) **4.** Describe the center and shape of a sampling distribution. (CLO 5) **5.** Explain the standard error as the variability of a statistic. (CLO 5) **6.** Predict how the sample size affects the standard error. (CLO 5)	e-Text: 3.1 Sampling Distributions Interactive Lecture: 3.1 Sampling Distributions Slides: 3.1 Sampling Distributions	Participation: Sampling Distributions – M&M activity: each student draws 10 M&Ms, computes $\hat{p}$ (blue), and adds it to a class dotplot; name the population, sample, and what each dot represents Your Turn: Parameter or Statistic?	Online Homework: 3.1 Sampling Distributions (due Sun Mar 7, 11:59 PM) Take-Home Quiz 4 (due Mon Mar 8, 11:59 PM) – covers P.1, 3.1–3.2 (all objectives) Test 2 (Fri Apr 2 – Sun Apr 4, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Apr 1) (all objectives)	WileyPLUS StatKey Blackboard Respondus
7. Construct a 95% confidence interval as statistic ± 2·SE, given the statistic and its standard error. (CLO 6) **8.** Interpret a confidence interval in context. (CLO 6) **9.** Explain the confidence level as the long-run proportion of intervals that capture the parameter. (CLO 4, CLO 6) **10.** Identify common misinterpretations of a confidence interval. (CLO 6)	e-Text: 3.2 Understanding and Interpreting Confidence Intervals Slides: 3.2 Understanding and Interpreting Confidence Intervals	Participation: Sampling Distribution (Kahoot!) – review of 3.1 Participation: Confidence Intervals – which of 20 simulated CIs capture μ; fix three misinterpretations Your Turn: Do These CIs Capture p?	Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Mar 7, 11:59 PM) Take-Home Quiz 4 (due Mon Mar 8, 11:59 PM) – covers P.1, 3.1–3.2 (all objectives) Test 2 (Fri Apr 2 – Sun Apr 4, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Apr 1) (all objectives)	WileyPLUS Blackboard Kahoot! Respondus

Also this week: Test 1 Retake – Mon Mar 1 – Sun Mar 7 (optional; higher score counts)